Characteristic Spectral Transform Distillation: Projector-Free Multi-Scale Geometry Matching for Language Models

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Abstract

Knowledge distillation for language models usually constrains output distributions while leaving intermediate representation geometry underspecified. Existing geometry-aware objectives often require learnable width-matching projectors, stored teacher prototypes, or explicit spectral decompositions. We introduce Characteristic Spectral Transform (CST) distillation, a projector-free objective that compares student and teacher hidden states through the multi-scale transform $\Phi_H(\gamma) = \log\det(I + \gamma G_H)$, where G_H is a centered, trace-normalized Gram matrix. Sampling several positive scales γ makes the loss sensitive to the complete normalized spectrum without directly matching eigenvalues or requiring equal hidden widths. A practical estimator uses a shared response-token subset, reuses one Gram matrix across scales, and evaluates all scales with batched Cholesky factorizations. This yields an auxiliary objective with no inference-time parameters and negligible measured kernel overhead. In a preliminary 10K-example distillation of Qwen2.5-14B-Instruct into Qwen2.5-1.5B-Instruct, CST improves the eight-task reasoning average from 44.6 for the reference base student to 49.9 under legacy-compatible metrics, while using only a fraction of the reference training budget. We also identify evaluation-version sensitivity on MATH and report both exact-string and symbolic-verification scores. These results motivate transform-level spectral supervision as a simple and scalable alternative to projector-based hidden-state alignment.

1 Introduction

Autoregressive knowledge distillation (KD) transfers behavior from a large teacher to a smaller student by aligning their predictive distributions (Hinton et al., 2015; Kim & Rush, 2016; Gu et al., 2024; Agarwal et al., 2024). Modern on-policy methods improve this transfer by training on student trajectories, reducing the mismatch between training and generation (Ko et al., 2024). Nevertheless, an output objective observes hidden representations only through the final vocabulary head. Many different intermediate geometries can therefore induce similar token distributions.

Matching hidden states directly is not an entirely satisfactory remedy. Teacher and student widths differ, making pointwise losses dependent on a learned projector whose parameters and inductive biases become part of the objective. Distributional summaries based on centroids add a precomputation stage and state that must be stored during training. Explicit singular-value or eigenvalue matching exposes the training path to expensive decompositions and unstable gradients around repeated eigenvalues. More fundamentally, matching a finite ordered spectrum treats the eigenvalues themselves as the representation, rather than comparing a characteristic functional of the geometry.

We propose CST distillation. For response-token hidden states $H \in \mathbb{R}^{m\times d}$, we center across tokens, normalize total energy, and form a positive semidefinite Gram matrix G_H with unit trace. We then evaluate

$$\Phi_H(\gamma) = \log\det(I + \gamma G_H), \qquad \gamma > 0, \tag{1}$$

at multiple randomly sampled scales. The student learns by matching the teacher's transform values at the same scales and token positions. The construction is invariant to translation, isotropic scaling, and orthogonal changes of hidden coordinates, and it naturally compares representations of different widths.

Our contributions are:

- We formulate hidden-state distillation as matching a multi-scale log-determinant transform of normalized geometry. The transform identifies the finite normalized spectrum over any nontrivial interval, but the training loss never computes an eigendecomposition.
- We give a width-agnostic, projector-free estimator using shared token and scale samples, the smaller determinant space, Gram reuse, and batched Cholesky factorization.
- We integrate CST into an on-policy output-KD pipeline and provide numerical, invariance, gradient, degeneracy, and dispatch tests. Profiling on an RTX PRO 6000 Blackwell reports approximately 2.2ms for the CST kernel at four layers, 64 tokens, and two scales.
- We report a transparent preliminary Qwen study, including the substantial discrepancy between legacy exact-string and modern symbolic MATH evaluation, and state the additional controlled experiments required for a definitive comparison.

2 Related Work

Language-model distillation. Classical KD minimizes a softened output divergence (Hinton et al., 2015), while sequence-level KD trains on teacher sequences (Kim & Rush, 2016). On-policy approaches instead expose the teacher to student-generated trajectories. MiniLLM uses reverse KL (Gu et al., 2024); GKD generalizes the divergence family (Agarwal et al., 2024); and DistiLLM combines adaptive sampling and skewed divergences (Ko et al., 2024). CST is complementary to these methods: it leaves output KD unchanged and adds intermediate geometric supervision.

Representation transfer. Intermediate-feature distillation has long used layer mappings and learned projections to reconcile dimensionality (Romero et al., 2015; Sun et al., 2019). Relational objectives transfer pairwise or higher-order structure instead of coordinates (Park et al., 2019; Tung & Mori, 2019). Recent spectral approaches diagnose or regularize representation collapse through rank, nuclear norm, or entropy summaries (Roy & Vetterli, 2007; Jing et al., 2022). Our distinction is operational and conceptual: CST matches a characteristic transform, uses neither a learnable width adapter nor a teacher prototype bank, and avoids explicit spectrum computation in training.

Log determinants. Log-determinant functionals occur in Gaussian information measures, experimental design, and diversity objectives. If p_i are the eigenvalues of a trace-normalized PSD matrix, Equation 1 equals $\sum_i \log(1+\gamma p_i)$. Rather than maximizing this quantity at a fixed scale, we use its curve over scale as a signature that the student matches to the teacher.

3 Characteristic Spectral Transform Distillation

3.1 Normalized hidden-state geometry

Let $H \in \mathbb{R}^{m\times d}$ contain the hidden states of m selected response tokens at one layer. Define

$$X = H - \mathbf{1}\bar{h}^\top, \qquad \bar{h} = \frac{1}{m}\sum_{i=1}^{m} H_{i,:}, \tag{2}$$

$$e_H = \|X\|_F^2, \qquad G_H = \frac{XX^\top}{e_H+\epsilon}. \tag{3}$$

For nondegenerate X, $G_H \succeq 0$ and $\mathrm{Tr}(G_H) \approx 1$. Centering removes global translation, while energy normalization removes isotropic scale. Because $XX^\top$ is unchanged by $X \mapsto XQ$ for orthogonal Q, the construction does not require teacher and student coordinates to align.

3.2 A multi-scale spectral signature

For *gamma* > 0, CST is Equation 1. Writing the positive eigenvalues of G_H as $p_1, \ldots, p_r$ gives the analysis identity

$$\Phi_H(\gamma) = \sum_{i=1}^{r} \log(1 + \gamma p_i). \tag{4}$$

Equation 4 explains the signal but is not the training algorithm. Small scales emphasize low-order spectral moments, whereas large scales increasingly expose rank and the distribution of energy among active directions. In particular,

$$\Phi_H(\gamma) = \gamma - \frac{\gamma^2}{2}\,\mathrm{Tr}(G_H^2) + O(\gamma^3), \tag{5}$$

because $\mathrm{Tr}(G_H) = 1$. The first nonconstant discriminative term therefore measures spectral concentration.

The transform is characteristic for a finite spectrum. Differentiating,

$$\Phi'_H(\gamma) = \sum_i \frac{p_i}{1 + \gamma p_i}, \tag{6}$$

a rational function whose poles and residues determine the nonzero p_i. Thus equality of transforms on an open positive interval implies equality of the normalized nonzero spectra, including multiplicities. Finite Monte Carlo sampling provides a stochastic approximation to this population criterion.

3.3 Distillation objective

For matched student and teacher layers ℓ_s and ℓ_t, we sample the same valid response-token indices and scales

$$\log \gamma_j \sim \mathcal{U}(\log \gamma_{\min}, \log \gamma_{\max}), \quad j = 1, \ldots, q. \tag{7}$$

The per-layer loss is

$$\mathcal{L}_{\mathrm{CST}}^{(\ell)} = \frac{1}{q} \sum_{j=1}^{q} \Big[\Phi_{H_s^{(\ell_s)}}(\gamma_j) - \mathrm{sg}\Big(\Phi_{H_t^{(\ell_t)}}(\gamma_j)\Big) \Big]^2, \tag{8}$$

and we average uniformly across n selected layers. The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{output-KD}} + \lambda_{\mathrm{CST}} \mathcal{L}_{\mathrm{CST}}. \tag{9}$$

No student-to-teacher hidden projector appears in this path. The teacher transform is computed under stop-gradient, and CST introduces no inference-time module.

3.4 Efficient and stable computation

We do not call a determinant, inverse, SVD, or eigendecomposition. Given the centered X, we construct only the smaller Gram space. When $m \le d$ we use $B = XX^\top/(e_H + \epsilon)$. If $d < m$, Sylvester's determinant theorem gives

$$\det(I_m + \gamma XX^\top/e_H) = \det(I_d + \gamma X^\top X/e_H), \tag{10}$$

so we instead use $B = X^\top X/(e_H + \epsilon)$. After symmetrization, all q matrices are constructed at once:

$$A_j = I + \gamma_j B, \qquad A \in \mathbb{R}^{q \times r \times r}, \quad r = \min(m, d). \tag{11}$$

A single batched Cholesky call produces $A_j = L_j L_j^\top$, followed by

$$\Phi_H(\gamma_j) = 2 \sum_k \log(L_j)_{kk}. \tag{12}$$

Table 1: Default CST training configuration. One token subset and one set of scales are shared by teacher, student, and all supervised layers.

Maximum response tokens m	64	Supervised layers n	4
Gamma samples q	2	Layer-depth range	$[0.20, 0.85]$
$\gamma_{\min}, \gamma_{\max}$	$10^{-2}, 10^{2}$	Sampling	log-uniform
Distance	squared ℓ_2	Arithmetic	float32

Table 2: Preliminary Qwen2.5-14B $\rightarrow$ 1.5B reasoning accuracy (%). Reference rows are contextual and use the NuNo-KD manuscript's full training budget. Our CST row uses 10K examples. MATH uses legacy-compatible exact match; math_verify for the same CST generations is 45.06.

Method	GSM8K	GSM+	MATH	MBPP	SciQ	STEM	Pro-Math	BBH	Avg.
Student (reference)	57.1	38.8	16.9	38.4	85.5	49.5	34.4	36.5	44.6
CSD (reference)	61.5	41.5	29.3	43.8	81.9	51.3	44.4	42.1	49.5
NuNo-KD (reference)	65.2	46.8	32.2	44.4	86.2	51.7	42.0	44.9	51.7
CST (ours, 10K)	65.58	46.79	14.92	46.80	86.90	51.95	44.49	41.87	49.91

All spectral arithmetic is float32. We retry with a small diagonal jitter only if Cholesky reports failure and skip layers with negligible centered energy. The dominant per-layer costs are $O(mdr)$ for the smaller Gram and $O(qr^3)$ for Cholesky, with $r = m = 64$ in our default LLM setting.

4 Experimental Setup

Models and data. Our preliminary experiment distills Qwen2.5-14B-Instruct into Qwen2.5-1.5B-Instruct (Qwen Team, 2024). Training uses 10K prompts from UltraInteract (Yuan et al., 2025), rather than the full roughly 78K budget used by the reference study. Student responses are generated on policy. We train for two epochs with global batch size 64, AdamW, learning rate 5×10^{-6}, cosine decay, 10% warmup, maximum length 1024, and output-KD temperature 1. Four uniformly spaced layers in relative depth $[0.20, 0.85]$ are supervised. Based on loss-scale calibration, we use $\lambda_{\text{CST}} = 0.003$; the raw transform loss has a different scale from nuclear-norm losses and should not inherit their coefficient.

Evaluation. We use lm-evaluation-harness with greedy decoding and the model chat template. The suite contains GSM8K, GSM-Plus, MATH, MBPP, SciQ, MMLU-STEM, MMLU-Pro-Math, and three-shot BBH-CoT. Evaluation runs in bfloat16 through vLLM on one RTX PRO 6000 Blackwell GPU. For compatibility with the published table we report the harness's exact-match MATH score in the main average. The current harness also provides symbolic math_verify; because it labels many mathematically equivalent answers that exact matching rejects, we report it separately and do not substitute it into the legacy average.

Baselines and reporting status. Student, CSD, and NuNo-KD numbers from the reference NuNo-KD manuscript (Anonymous, 2026) are included only as contextual references, not as controlled reproductions. Their training budget and, for MATH, evaluator version differ from ours. A final submission must replace these comparisons with checkpoints trained on the same data and evaluated by the same frozen harness. We include this explicit limitation to prevent the preliminary results from being interpreted as a definitive ranking.

5 Results

Despite using 10K rather than roughly 78K prompts, CST improves the reference base-student average by 5.3 points and reaches 49.91 under the legacy-compatible metric selection

(Table 2). Its GSM8K score is 65.58, with standard error 1.31 points, statistically indistinguishable from the reference NuNo-KD value of 65.2. Gains are also visible on MBPP, SciQ, MMLU-STEM, and MMLU-Pro-Math. BBH-CoT remains below the reference NuNo-KD score.

MATH requires special care. Current-harness exact match is 14.92, whereas symbolic verification gives 45.06 on identical generations. Among 4,454 available per-sample records, 1,409 are rejected by exact match but accepted by symbolic verification; inspection shows many contain a correct boxed answer without the exact expected “Final Answer” surface form. Only four examples show the reverse disagreement. This establishes strong evaluator sensitivity but does not establish that 45.06 is comparable to the published 32.2. The scientifically valid resolution is to evaluate every checkpoint using both frozen metrics.

Efficiency. At $m = 64$, $q = 2$, and $n = 4$, the reported CST transform computation is approximately 2.2ms, while an optimizer step in the complete on-policy KD pipeline takes roughly 22.6s. This timing isolates the transform kernel and does not imply that hidden-state capture is free; nevertheless, it indicates that batched log-determinant evaluation is not the runtime bottleneck. The model uses about 31.6GiB allocated GPU memory after initialization in the reported setup and encounters no CST-related out-of-memory event.

6 Analysis and Discussion

6.1 Why multiple scales?

At a single small γ, unit trace makes transforms nearly identical. At one large γ, the signal can be dominated by effective rank and very small eigenvalues. Log-uniform sampling over four orders of magnitude mixes these regimes without constructing a dense grid. Crucially, teacher and student receive identical samples, so stochasticity does not introduce a scale mismatch. An ablation over $q \in \{1, 2, 4\}$ and narrower scale intervals is still needed to quantify this design choice.

6.2 Relation to rank-profile matching

Rank-profile matching explicitly eigendecomposes a normalized Gram matrix, sorts eigenvalues, and compares cumulative spectra. It is a useful analysis and baseline tool, but it is not CST. CST compares evaluations of an analytic transform and obtains gradients through Cholesky factorization. It therefore avoids eigenvector sensitivity at repeated eigenvalues and does not turn the method into direct eigenvalue regression.

6.3 Loss-scale calibration

Raw CST losses in early training are commonly tens of units because the squared gap aggregates transform differences across broad random scales. A coefficient copied from a differently normalized representation objective can therefore dominate output KD. In our run, $\lambda_{\text{CST}} = 0.2$ would contribute roughly 12 units when output KD is about 1.4. We instead select 0.003, yielding a typical contribution of 8–16% of output loss. Future work should tune this value on a held-out validation protocol or normalize the distance while preserving the transform-level objective.

7 Limitations and Required Confirmatory Experiments

The current evidence is preliminary. First, the 10K training budget differs from the full reference budget, and published baseline numbers are not same-run controls. Second, we report one seed and one model pair. Third, MATH metrics changed across harness versions. Fourth, the present objective samples response tokens globally across a batch; while indices are shared across models and layers, alternative per-sequence sampling may change the geometry. Finally, transform equality is identifying over a continuum, while training observes only two random scales per step.

A confirmatory study should include: (i) full-budget Qwen and Gemma2 model pairs; (ii) output-KD, NuNo-KD, rank-profile, and CST trained with identical seeds and data; (iii) at least three seeds for the principal comparison; (iv) ablations over $m \in \{32, 64, 128\}$, $q \in \{1, 2, 4\}$, $n \in \{2, 4\}$, gamma range, centering, and loss weight; (v) wall-clock and peak-memory comparisons including hidden-state capture; and (vi) frozen evaluation containers with both exact and symbolic MATH scoring.

8 Conclusion

We introduced CST, a projector-free representation-distillation objective based on the characteristic curve $\log\det(I + \gamma G_H)$. Matching this curve at shared random scales transfers normalized spectral geometry without matching coordinates, widths, or eigenvalues directly. The estimator uses small response-token subsets and batched Cholesky factorizations, making its measured kernel cost negligible within on-policy LLM distillation. A preliminary 10K-example Qwen experiment is promising but also exposes the need for controlled full-budget baselines and frozen MATH evaluation. We view these transparent intermediate findings as evidence that transform-level geometry is a useful direction, rather than as a final claim of superiority.

Ethics Statement

This work studies model compression and may reduce inference resource use. Distillation can also transfer teacher errors, biases, and unsafe behaviors; the proposed representation objective does not mitigate these risks by itself. Benchmark accuracy should not be interpreted as evidence of safety or fitness for deployment.

Reproducibility Statement

The implementation exposes token count, layer range, gamma distribution, number of scale samples, distance, centering, jitter, and loss weight. Tests cover Cholesky–log-determinant agreement, Sylvester's identity, invariances, shared gamma samples, unequal hidden widths, teacher stop-gradient, degenerate representations, and absence of projector/NuNo state in dispatch. Exact commands and evaluator outputs should accompany a submission; the present draft marks results that still require controlled reproduction.

AI Use Statement

Generative AI tools assisted with code review, experiment-log summarization, and preparation of an initial manuscript draft. The authors are responsible for verifying all mathematical statements, citations, experimental results, and final prose, and for ensuring compliance with the conference policy.

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A Reference implementation pseudocode

```
indices = sample_response_tokens(labels, max_tokens=m)
gamma = exp(uniform(log(gamma_min), log(gamma_max), [q]))
losses = []
for student_layer, teacher_layer in matched_layers:
    Xs = center(student_hidden[student_layer][indices]).float()
    Xt = center(teacher_hidden[teacher_layer][indices]).float().detach()
    Bs = smaller_gram(Xs) / (squared_frobenius(Xs) + eps)
    Bt = smaller_gram(Xt) / (squared_frobenius(Xt) + eps)
    As = I[None] + gamma[:, None, None] * Bs[None]
    At = I[None] + gamma[:, None, None] * Bt[None]
    phi_s = 2 * log(diag(cholesky(As))).sum(-1)
    phi_t = 2 * log(diag(cholesky(At))).sum(-1)
    losses.append(mean((phi_s - stopgrad(phi_t)) ** 2))
return mean(losses)
```

B Additional numerical safeguards

Hidden states with centered energy below ϵ are skipped. Gram matrices are explicitly symmetrized to suppress floating-point asymmetry. We use cholesky_ex to inspect factorization status and add $10^{-6}I$ only on failure. Teacher hidden states and transforms are detached. All selected layers share response-token indices and gamma probes within an optimization step.

C Per-category MATH scores

Current-harness symbolic-verification accuracies are 63.77 (algebra), 37.55 (counting and probability), 38.20 (geometry), 26.69 (intermediate algebra), 35.74 (number theory), 60.39 (prealgebra), and 32.05 (precalculus), for an aggregate of 45.06. Corresponding exact-string scores aggregate to 14.92. These values are reported to make evaluator-version effects auditable.